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Environmental risk and deliberative planning in the Locarnese region of Switzerland

Annette M. Bardsley[a,*], Douglas K. Bardsley[a], Marco Conedera[b], Marco Pütz[c], Guy M. Robinson[a] and Delene Weber[d]

[a] Geography, Environment and Population, School of Social Sciences, The University of Adelaide, Adelaide, South Australia, Australia
[b] Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Cadenazzo, Ticino, Switzerland
[c] Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Birmensdorf, Zurich, Switzerland
[d] UniSA STEM, The University of South Australia, Mawson Lakes Campus, Australia
[*] Corresponding author, email: annette.bardsley@adelaide.edu.au

Declaration
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Highlights

- The nature of wildfire risk is changing in the southern Switzerland
- Local environmental risks are not rated highly by household respondents
- Residents state that they have a very high understanding of the planning system
- Many respondents do not feel that they can influence planning outcomes
- Highly deliberative planning systems do not necessarily reflect local opinions

Keywords: deliberative democracy, planning, wildfire, perceptions, peri-urban, Switzerland

Abstract

A survey of residents in the Locarnese region of Canton Ticino, Switzerland examined perceptions of both increases in exposure to changing levels of environmental risk and the deliberative Swiss planning system. In the context of a growing risk of wildfires in the study region, residents in the peri-urban residential areas of the forested south-facing slopes of the Swiss Alps were asked a series of questions regarding their relationships to the planning system, place and environment, and attitudes to risk. Descriptive and inferential statistics revealed high appreciation of lifestyle and amenity values and support for firefighting services. While survey respondents did not consider that those risks are increasing with environmental change, wildfires and landslides were recognised as posing significant risk. Current levels of urban expansion were viewed as excessive, heightening planning challenges at the interface between forests and settlements. Although the deliberative governance arrangement offers citizens with the opportunity to be involved in decision-making through official channels, many respondents felt their views were inadequately reflected in decisions and that key issues are not being addressed by the planning system. In contrast to other respondents, those who are younger, of local background or less educated are more dissatisfied with their influence over planning, which could have important implications for hazard management as new risks emerge with environmental change. If expectations of involvement are not met, new challenges will be presented to the distinctive Swiss planning system.

1. Introduction

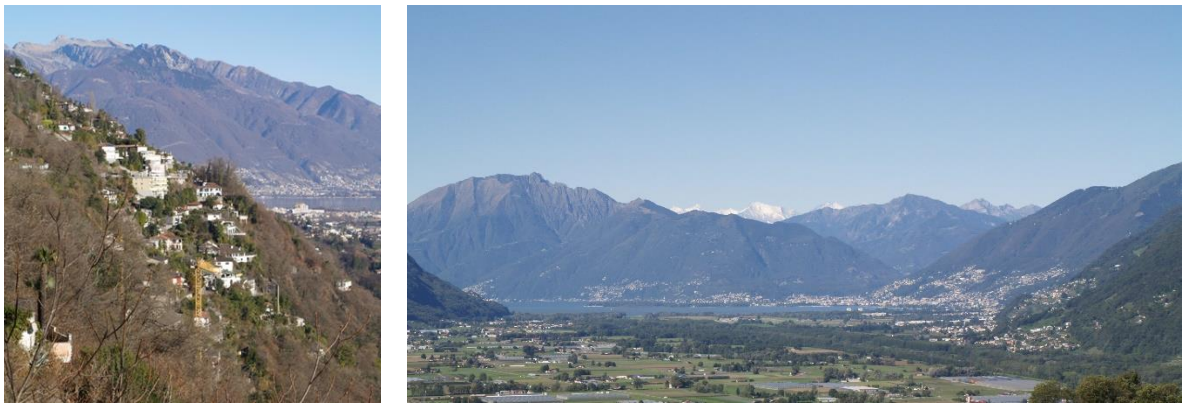
Global climate change is shifting many of the dynamic relationships between the environment and society. One of the major implications of that change is the increasing likelihood of wildfires, even in places that have not traditionally considered themselves at high wildfire risk, such as the Swiss Alps (Valese *et al.* 2014; Conedera *et al.* 2018). In the alpine landscape, wildfire risk is also being exacerbated by factors such as reafforestation and urban development. In the Swiss Italian Canton of Ticino, the forested area has more than doubled in the last 100 years, to cover 52 percent of the canton in 2018 (Galfetti 2020). At the beginning of the 20th century, the forest cover in Ticino was 60,000 hectares (ha), but by 1950 had grown to 85,000 ha, and by the beginning of the 21st century had reached 140,000 ha (USTAT 2007). This increase in forest cover has been particularly evident at altitudes greater than 800 metres above sea level (Bettelini 2007), where previously managed agrarian and pastoral landscapes have transitioned to forest (Bauer *et al.* 2009; Mack *et al.* 2013; USTAT 2013; Loran *et al.* 2017). At lower altitudes, the forests have also expanded into agricultural lands and are increasingly in close contact with settlements (Galfetti 2020). The positive outcomes of that expanded forest area are significant improvements in erosion control (Rickli and Graf 2009; Malek *et al.* 2015), and, for many people, biodiversity conservation and amenity values through the rewilding of landscape (Pereira 2015; Prior and Brady 2017; Frick *et al.* 2018). Nevertheless, the change has also generated different forms of environmental risk. While forests have largely been perceived historically as a means of mitigating the risk of natural hazards (Mather and Fairbairn 2000), as the forest has expanded in areas close to townships, greater risks of wildfires and tree fall have emerged in peri-urban areas (wildland-urban interface) (Von der Dunk *et al.* 2011; FOEN 2014; Conedera *et al.* 2015).

Concurrently, increased population growth rates in the second half of the 20th century in Canton Ticino, and the associated decline in availability of residential land on valley floors, have changed the traditional urban settlement patterns that had dominated since the Middle Ages (OST-TI 2014). In the Locarnese region of Ticino, this expansion has been particularly evident since the 1970s, and has come at the expense of terraced vineyards and grazing land that acted as buffer zones between the forest and the village. Traditionally, a trading centre between Switzerland and northern Italy, high-level industrial activities and sun-seeking tourists and retirees have supported growth in the residential population in the Locarnese region from 21,031 in 1950 to 38,398 in 2000 (USTAT 2020). Since 1985, agricultural areas have diminished by 20 percent across Ticino (Galfetti 2020), and the loss of traditional landscapes has become a sensitive cultural issue (Rodewald *et al.* 2014), as

suggested by the high level of cantonal support for a proposal seeking to curb urban sprawl in Switzerland during a plebiscite in February 2019 and the initiative to protect cultural landscapes (Geiser 2019; Iniziativa Paesaggio 2020).

The key research aim is to examine how the highly deliberative Swiss planning system is managing changes in environmental risk. As the landscape of Canton Ticino changes and residential areas are brought into direct contact with the expanding forest (Figure 1), this paper analyses how residents of the steep peri-urban areas around Locarno perceive of the environment and changing natural hazards in association with opportunities for appropriate planned responses. The analysis is particularly relevant given that the highly deliberative Swiss political system aims to generate planning outcomes that reflect people's perceptions of landscape and environmental risks. The paper appraises the potential for participatory planning strategies to provide the community with opportunities to express their attitudes about changing levels of environmental risk, and thereby influence effective planning in the Locarnese peri-urban area.

Figure 1: The Locarnese region of Canton Ticino, Switzerland (2015), indicating how the forest interacts with settlements



2. Deliberative planning for risk

Risk is a complex term defined as a 'combination of the probability of an event and its negative consequences' (UNISDR 2007). New levels of environmental risk associated with climate change are driving societies accustomed to dealing with natural hazards to acknowledge the potential limits of current disaster risk governance and management capabilities (Kanonier and Rudolf-Miklau 2018; Bardsley and Knierim 2020). This situation reflects a structural crisis identified by Beck (1986), whereby established modern management frameworks no longer adequately represent the fundamental shift that has occurred in the parameters defining risk across society. Beck's theory

considers the significant complexity of the current social reality and notes that first modernity approaches to risk management have focussed on reducing the impacts of natural disasters through technological advancement. Yet, with increasing risk, boundaries or thresholds are reached after which consequences and likelihoods continue to escalate irrespective of substantial further investments in physical management interventions.

Beck *et al.* (2003) evoke a new, second or reflexive modernity, in which governance must work to incorporate an understanding of risk into all contemporary policy and planning. Such a socialisation of risk builds upon Habermas' (1989) reflections on the need for processes or 'social spaces' that allow citizens to engage and participate in decision-making (Parkins and Mitchell 2005). Together these conceptions inform the fundamental premise of this paper, namely that landscape planning must become more reflexive to respond to the unique situations that are presented locally to manage environmental risk (Godschalk *et al.* 2003; Morehouse *et al.* 2011; Von der Dunk *et al.* 2011; Bardsley *et al.* 2015; Sharma-Wallace 2016). In recent years, there has been renewed interest by planners in the advantages of early and ongoing community engagement to educate and understand people's knowledge, values and preferences for effective land-use management that progresses beyond simplistic consensus-based outcomes (Healey 1997; Forester 1999; Heberlein 2012; Gallent and Ciaffi 2014; Legacy 2017; Natarajan 2017). Yet, such an approach requires community understanding of environmental planning systems and effective engagement processes to influence spatial and systemic outcomes.

How people perceive of risk is hugely influential over their level of support for particular planning outcomes. Issues of transparency and meaningful public participation and engagement are a developing research field, with numerous studies focussing on the mitigation of wildfire risk (Martin et al. 2009; Frandsen *et al.* 2011; McGee 2011; Eriksen 2014; Ghasemi et al. 2020). In conjunction with public entities, such as biodiversity conservation and emergency response agencies, local planning bodies have the complex role of translating government policies into tangible procedural and spatial outcomes aimed at minimising the risk factors associated with natural hazards, while supporting social values. For that reason, the social and cultural influences over perceptions and responses to risk are a vital component of environmental risk research (Pfister 1985; Slovic 1987; Pfister 2009; Xue *et al.* 2014; Reid and Beilin 2015). For example, research on individual resident and community-preparedness for wildfires has identified significant links between risk minimisation and residents' place values (Anton and Lawrence 2014; Frick *et al.* 2018; Weber *et al.* 2019; Koksal et al. 2019). The argument runs that as risks of wildfire increase, approaches to manage the forest in

association with urban expansion will need to adjust by taking into account both the new science and stakeholder perceptions of risk.

3. The Swiss deliberative democratic system

Environmental planning relies on a range of different spatial, landscape and natural hazard planning instruments. The challenge lies in identifying the tools and approaches to achieve decisions that are both reflective of local needs and broader policy outcomes (Healey 2006). Apart from the fact that several of the authors are Swiss, Switzerland was chosen as a case-study area because, whilst many countries make use of varying levels of deliberation to engage their population in decision-making, no other jurisdiction has such entrenched participatory mechanisms (Steiner 2012; Hill 2013; Della Porta et al. 2017). Dryzek (2009) defines the deliberative capacity of a political system as the degree to which decisions and then outcomes truly reflect the wishes and values of the population. Both representative and direct democratic institutions guide the deliberative Swiss democracy (Parkinson 2006; Pedrini 2014).

The Swiss federal system entails three governance levels: federal, cantonal and municipal. Although the legislative framework for spatial planning has been empowered at the federal level since 1969 (Lendi 2012), in 2012, the three levels of government collaborated to develop and ratify 'Concept Switzerland 2012', to provide nation-wide strategic planning guidelines (Swiss Federal Council 2012). Through the principle of municipal federalism, local authorities are granted wide-ranging powers in relation to land use and the issuing of building permits. While the strategic and legislative planning frameworks work across all levels of governance, much of the implementation is left to the 26 Cantons and the over 2000 municipalities to develop detailed structured and land use plans (Grams and Nebel 2013). In practical terms, while the responsibility for transport infrastructure, economic policy and environmental protection fall to the federal government, governance of land use planning and nature conservation falls mostly to the Cantons (Muggli 2013). Nevertheless, the Swiss federal government reserves the right to regulate key areas as defined in the constitution, by approving cantonal plans and ensuring key principles are supported, such as the separation of building and non-building zones.

In the Swiss model, citizens have direct access to the complex decision-making system through referenda and popular initiatives. Partly for that reason, Switzerland has a planning system that enjoys a rare level of public support and respect (Lendi 2012; Muggli 2013). The direct democratic

principles apply to all levels of government, and while the items voted on are often high-level legislative frameworks and planning principles, at cantonal and municipal levels citizens can directly influence local outcomes. The processes of participation have promoted innovative outcomes, such as the establishment of sustainable mountain forestry (Hersperger *et al.* 2017) and strict controls on urban sprawl (Gennaio *et al.* 2009; Swiss Federal Council 2012; Klaus 2021). In recent times several intergenerational issues have been shaped by popular interventions at the cantonal level in Ticino (Repubblica e Cantone Ticino 2021). In 2007, following the successful collection of signatures in the Canton, the direct intervention of a citizen referendum overturned a parliamentary decision to build a freeway leading to Locarno across the Magadino plain. In another example, citizen actions ensured that existing cantonal laws to safeguard natural and cultural local heritage were updated to help sustain traditional landscapes.

The successful Swiss spatial planning system has emerged from a traditional culture accustomed to dealing with environmental risk, which enabled the nation-state to thrive despite the constraints of an extreme landscape and hazardous terrain (Bründl *et al.* 2009; Pfister 2009). Historically, the Swiss people have been characterised by a strong sense of cultural attachment to place, and of shared responsibility in the governance of the country (Pfister 1985; Kriesi and Trechsel 2008). While participatory mechanisms in the governance process aim to provide the Swiss with a deeply encultured understanding of risk mitigation, that assumption has rarely been tested by research on environmental hazards (Maidl et al. 2021). Implicit to the approach is the assumption that the opportunity to be heard by authorities leads to citizens feeling that they are valued as actors in the development of new understandings, decisions and institutional capacities.

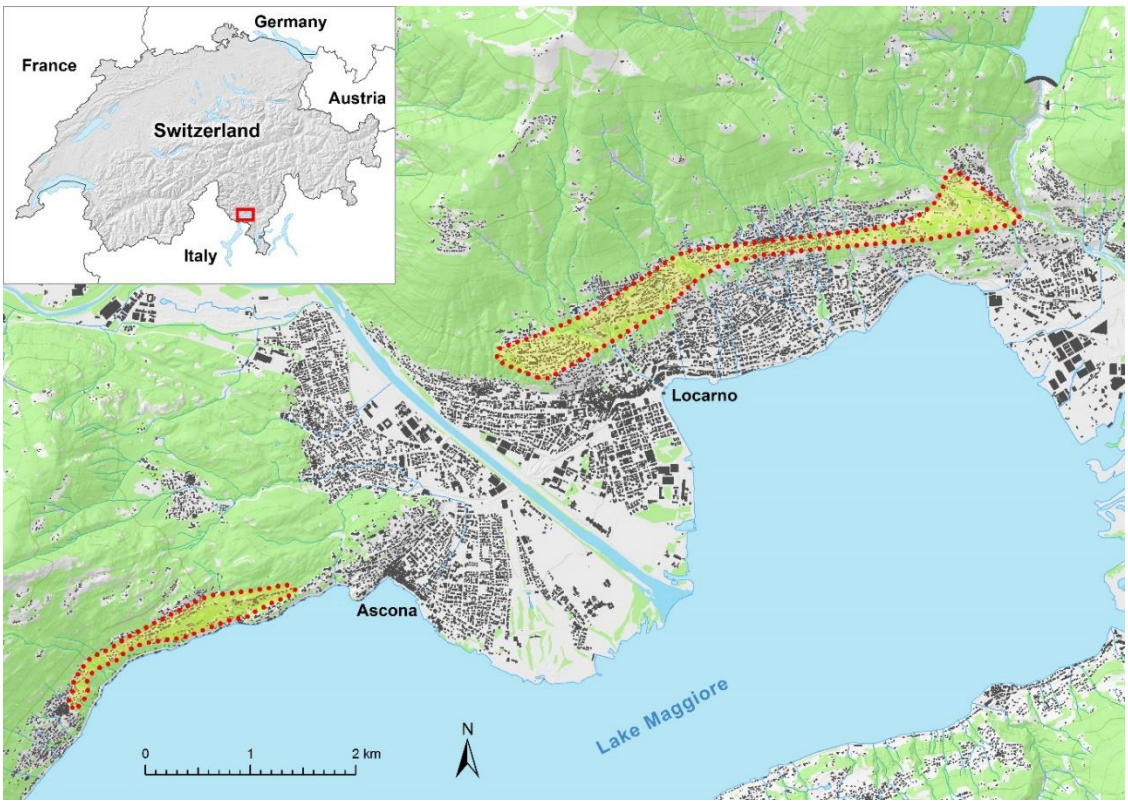
One of the specific challenges confronting the inclusive Swiss decision-making system is for citizens to feel that local spatial outcomes truly reflect the values of the community. If citizens feel that they do not have a voice in planning issues, they may feel disaffected and distrustful (Bauer et al. 2014), or on the other hand, simply become increasingly reliant on expert technical interventions (Turner 2001). Recently, the Swiss government has identified a specific lack of understanding amongst the general population of the potential effects of increasing environmental risks (Camenzind and Loat 2014; FOEN 2016). While opportunities for uniquely high levels of deliberation and engagement exist in Switzerland, if the general population is not informed about, is unaware of or holds false perceptions of risk, deliberative democracy could lead to undesirable planning outcomes. Therefore, both the opportunities presented by citizen participation and the risks of the approach generate important examples for policy development in other countries as they also struggle to manage risk

(eg. Smith 2009). Swiss federal and cantonal authorities are now acknowledging the necessity for improved communication and community education on the new types of risk (FOEN 2016). It is within such a context that we examine how residents of peri-urban Locarno perceive of environmental risks and their influence over planning decisions.

4. Wildfire risk in the study area around Locarno

The research was undertaken in the peri-urban areas of Locarno, within the municipalities of Locarno-Bré, Orselina, Muralto, Minusio, Brione sopra Minusio, Minusio and Ronco sopra Ascona (Figure 2).

Figure 2: Household Survey areas (2015) within the municipalities of Locarno-Brè, Orselina, Muralto, Minusio, Brione sopra Minusio, Minusio and Ronco sopra Ascona



The Locarnese region is located in the southern foothills of the European Alps in the Swiss-Italian Canton of Ticino. In broad terms, the local climate is defined as Warm Temperate, with warm, wet summers and cool to mild winters, but is also characterised by highly variable temperatures and precipitation (Spinedi and Isotta 2004; MeteoSwiss 2012). The area averages 1457 millimetres of annual rainfall and has the highest average temperatures for the country. The slopes of the

Locarnese region are covered in a mixed-forest, dominated at lower altitudes by sweet chestnut (*Castanea sativa*), a tree rich with cultural significance and a forest-type that is traditionally more likely to burn during the late-winter dormancy phase (Bajocco *et al.* 2011).

Palaeoecological studies provide evidence of frequent wildfires in southern Switzerland since pre-historic times (Tinner *et al.* 1999). The forests of Canton Ticino were severely depleted in the nineteenth century, leading to slopes denuded of trees (Pezzatti *et al.* 2013). Replanting activities at the turn of the twentieth century and forest expansion following the post-WWII abandonment of marginal agricultural areas has led to a significative increase in forest area (Bettelini 2007; Loran *et al.* 2017). With more forest there has been an associated increase in wildfire events, but that trend was successfully reversed through policy interventions, such as the 1987 Cantonal Executive Decree banning private burn offs, and increased investment into fire-fighting infrastructure (Conedera *et al.* 2004; Sezione Forestale 2008; Pezzatti *et al.* 2013). Now however, wildfire risk is projected to increase with a changing climate, forest conditions and settlement patterns (Zumbrunnen et al. 2011; Pezzatti et al. 2016), and the Federal Office for the Environment (FOEN) has called for Cantons to have a greater consideration of forest fire mitigation strategies (FOEN 2016, p.39).

Long-range climate forecasts suggest that Canton Ticino will experience an increase in the number of hot days and higher median temperatures across all seasons, while summer precipitation is projected to decrease and winter precipitation to increase (Spinedi and Isotta, 2004; MeteoSwiss 2012; Moser and Del Priore, 2013; FOEN 2017). Over the last thirty years, temperatures in Ticino have risen at a rate almost double the median global average (MeteoSwiss 2012). Already, over the period 1880 to 2005, the length of summer heatwaves over Western Europe, including Switzerland, has doubled on average and the frequency of hot days has almost tripled (Brönnimann *et al.* 2014). Although rainfall has always been variable, an increase in the number of dry spells has become apparent since the 1970s (Rebetez 1999; Reinhard *et al.* 2005). This tendency, along with the overall warming trend in the Alpine region, is partly why concerns are being raised that environmental conditions are becoming more favourable for dangerous wildfires and other hazards (Rebetez 1999; Adger 2006; Wastl *et al.* 2013; Pezzatti *et al.* 2016; Bottinelli 2017; Matasci *et al*. 2018). Those changes have the potential to adversely affect the protective and recreational functions of the forest and increase infrastructure damage (FOEN 2014; IFEC Ingegneria SA *et al.* 2016).

Concern about the likely impacts of droughts and wildfires in Ticino continue to grow amongst the scientific community, especially as they might generate risks with which people are less familiar

(Reinhard *et al.* 2005; Rigling and Schaffer 2015; PLANAT 2018). Some of the most extensive wildfires experienced in Switzerland occurred during the 2016-2017 winter in Ticino and neighbouring Grisons (Pezzatti et al. 2017; Degrussa 2018). Although there was no loss of human life, the significant destruction of forests established to protect settlements, exposed communities and infrastructure to avalanches, landslides and debris flows in disturbed landscapes (Conedera *et al.* 2003; Alpine Initiative 2018). As risks increase, the conception of wildfires in the region has the potential to change from a regular, although manageable issue, to a situation where new, more extreme forms of the hazard will become apparent (Henne Bigalke *et al.* 2018). Reinhard *et al.* (2005, p.8) conclude for Switzerland, 'Decision makers have to deal with more severe framework conditions than twenty or thirty years ago and should implement appropriate measures'. To do so, more specific knowledge is required on how authorities can draw from local knowledge to mitigate risk.

5. Method

Both quantitative and qualitative data on residents' perceptions in the Locarnese area were generated through a residential household survey undertaken in October-December 2015. The survey was conducted to identify perceptions of environmental values, as well as attitudes towards the local planning system's management of those values and risks. Participants were sampled using stratified selection criteria (Matthews and Ross 2014; Etikan *et al.* 2016), including: primary residence; proximity to naturally vegetated areas; and, exposure to geophysical hazards associated with steep slopes (Figure 2). Addresses were obtained from the Population and Migration Office of Canton Ticino, which listed addresses by a 'head of household' criteria, and in most cases, identified an adult male as the household representative. To attempt to balance a potential gender bias, where multiple residents appeared under the same address, or where two individuals (one male and one female) were listed as heads of household for an address entry, a female addressee was selected. Households included within the village cores or '*nuclei*', of the six municipalities were excluded from the sample due to lower levels of environmental risk.

The survey contained Likert-scale, nominal binary (yes, no) and open-ended questions to produce both quantitative and qualitative data. Key questions asked about residents':

- gender, age, household typology, nationality, place of birth, employment and education, home (length of stay, previous place of residence, factors influential in the choice of location and on possible relocation, location in relation to forest and green spaces);
- location (most and least appreciated aspects, attachment levels);
- perceptions of forest management and biodiversity conservation;

- perceptions of environmental risk factors and risk management;
- actions and behaviour to prepare for environmental hazards and protect the environment;
- perceptions of the efficacy of agencies responsible for fire-fighting;
- opinions about and knowledge of the planning system; and
- satisfaction levels with planning decisions and engagement processes.

To optimise coverage in this multi-lingual context, a mixed-mode multi-language survey was applied. The paper questionnaire was mailed to residents in the main local language, Italian. However, links to online Survey Monkey versions of the questionnaire were offered in both German and English. Surveys were posted through the Swiss postal service in October 2015 to 900 households, with reminder postcards sent out in late November 2015. A total of 146 valid responses were received (paper surveys 124 and online surveys 22), providing a response rate of 16 percent. That response rate is in line with similar surveys conducted in Switzerland ranging from 15 percent (Home et al. 2014; Maidl et al. 2021) to 30-35 percent (Conedera et al. 2015; Frick et al. 2015). While the response rate is low and the results here are indicative only, they provide important insights into the ways that the Swiss respondents perceive of their environment and planning systems.

To guide the data analysis, responses were organised into three themes linked to the broader research objectives: conceptions of place and environment; attitudes to risk; and the attitudes towards the planning system. The strength of associations between the variables within the three thematic groups were analysed using Pearson Chi-Square (χ^2) and Fishers Exact Tests (FET) in IBM SPSS (version 22). While a large body of information was generated from the survey, specific data are chosen and presented here to develop narratives on perceptions of environment and management. In association, qualitative data generated through open-ended questions in the survey provide greater depth of insight into the perceptions of respondents, and quotations are used to provide in-text support for key arguments.

6. Results

6.1 Characteristics of survey respondents

A total of 146 valid surveys were received, although not all questions were answered in each case (Table 1). Of those respondents who answered, 90 percent indicated that they held Swiss citizenship and more than half (54.8%) were born in Canton Ticino, with nearly a third (31.9%) stating another Swiss canton as their birthplace and the remainder (13.3%) born abroad. Over one third of

respondents (37%) had lived at their current address for 21 years or longer; 25 percent had lived at the address for 11-20 years; and 38 percent had moved into the area in the last 10 years, thus revealing a relatively sedentary sample. This permanency could imply a good knowledge of the location and associated environmental risks. Yet, it may also infer old age and perhaps a greater sense of vulnerability, an assumption supported by the finding that 43 percent of respondents were aged 65 years or more. That cohort are also likely to be less-affluent than the broader population and dependent on the aged-pension (Swiss Life 2015). Nevertheless, cantonal data for the surveyed municipalities reveal an average of 56 percent of all residents aged from 18 to 64 years - very close to the 57 percent in that age bracket within the sample (USTAT 2016). Therefore, while the response rate is quite low, it is relatively representative in terms of age. Some 65.8 percent of the total 146 participants were male and 34.2 percent female, whereas data for the six municipalities show a slight discrepancy, with a predominance of females (53.1%) over males (46.9%) (USTAT 2016).

Table 1. Characteristics of respondent survey sample (N=146)

Characteristics	Frequency (n)	Groups	Percent
Age	146	18-44	19.2
		45-64	37.7
		65+ years	43.2
Gender	146	Female	34.2
		Male	65.8
Time lived at same address	146	0-10	38.4
		11-20	24.7
		21+ years	37.0
Previous postcode	125	Ticino	71.2
		Other	28.8
Education level	138	Secondary	52.9
		Tertiary and higher	47.1
Household type	140	With child/ren	30.0
		Two-person	45.7
		One-person	20.7
		Other	3.6
Home ownership	145	Owners	64
		Renters	36
Secondary home or *Monte** ownership	144	Yes	36.1
		No	63.9

* Traditional transhumance residence

6.2 Respondents' perceptions of place and the environment

Household respondents expressed high appreciation for the local landscape, emphasising peace and quiet, proximity to nature, views, clean air and a relaxed lifestyle. Green spaces were perceived as

particularly important (Table 2), with lifestyle and hedonistic values, including pleasant visual landscapes and favourable climate, also highly rated.

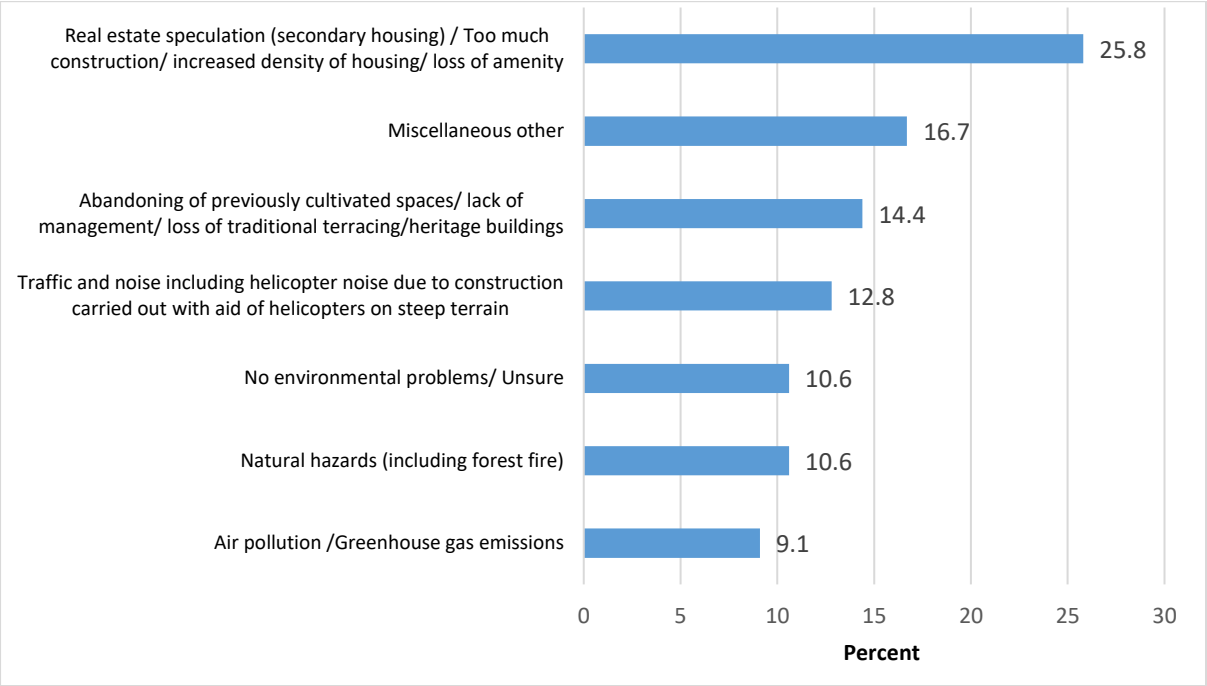
Table 2: Respondents' perceptions of the Locarnese region

	Percent		
Personal values	**Agree and strongly agree**	**Neutral**	**Strongly disagree and disagree**
The green spaces in the Locarnese are very important to me (n=138)	91	7	2
I am very attached to the Locarnese region (n=139)	83	15	2
The more people that live in Ticino, the less satisfied I am (n=136)	41	32	27
The municipal zoning plan supports conservation of natural assets (n=138)	30	28	43
Vineyards are more important than forests (n=137)	20	38	43
Prioritise urban development over nature conservation (n=139)	10	12	78

Respondents were asked why they chose to live in their current location. The standout motivations were peace and quiet (85%), the view (80%); and proximity to nature (80%). Other important factors included: relaxed lifestyle (77%); clean air (74%); escape from urban life (70%); and a good environment to bring up children (67%). These results highlight the 'Swiss sunroom' status of the Locarnese region and help to explain the attraction of its mild climate and environment. Some 94 percent of all respondents agreed with the statement that forests significantly contributed to the beauty of regional landscapes.

Usage of recreational areas (green spaces, forests, parks) was very high among respondents, with 71 percent using them at least once a week. Only approximately one-third of respondents indicated proximity to work and transport/services as being important in their location decisions. While people were concerned about the degradation of the environment and urban development, factors impacted by climate change such as wildfires did not rate highly amongst respondents in relation to other factors (Figure 3).

Figure 3: Percentage of respondents' perceptions of the most significant environmental issues for their municipality (n=105)



Respondents were provided with a list of six common hazards and asked to indicate which they felt posed the major threat to their home. When evaluating those items, we see that wildfires and landslides were the greatest concern to residents living on the slopes around Locarno (Table 3).

Table 3: Most important major natural hazard threatening the home of respondents (n=120)

Natural hazard	Number	Percent
Wildfires	61	34.3
Landslides	50	28.1
Rockfalls	39	21.9
Flooding	14	7.9
Earthquakes	12	6.7
Avalanches	2	1.2
	178*	100

*Multiple responses allowed

Over half of the respondents indicated having some personal experience of wildfires in Ticino, with 18 respondents providing detailed examples of their experiences with wildfires, including childhood memories of watching the blazes on the hillsides from afar; with only two respondents indicating that they had either their house or adjacent buildings damaged or destroyed during a wildfire.

- One female resident aged between 40-44 years commented: '*I remember the large fires on the mountains in my childhood, when you would be fascinated by the colour and the smoke plumes [of the fires].'*
- A male resident aged over 65 years said: *'In the 1950-70s, the hills were often on fire. I remember the death of a fire-fighter during my military training, the father of a school mate.'*

The quantitative data also show that those respondents who consider wildfires as the principal environmental threat are more likely to have knowledge of previous wildfire events near their home (FET (n=99) = 12.160, p = .002). That result could in part be explained by the fact that respondents who were born in Ticino rated the threat of forest fires to their home higher than those respondents born elsewhere (χ^2 (n=102) = 4.965, p = .042). No statistically significant difference appeared in the relationship between age and those who consider wildfires as a major threat.

Respondents were asked to select the most likely hypothetical timeframe for the occurrence of a major wildfire occurring in their vicinity. A substantial 51 percent of respondents indicated that they were 'unsure' about the timeframe surrounding a possible wildfire and were unwilling to speculate. Results for the remaining 49% of respondents (n=68) showed that most (69.1%) thought that a wildfire was likely to occur within ten years. Although not statistically significant, younger respondents were more likely to think that such as fire was unlikely to occur for a long time (Table 4). Interestingly, 16.2 percent of respondents felt that a wildfire was never likely to occur, with almost one-quarter of young adults answering in that manner.

Table 4: Percentage of perceived likelihoods of a major fire by age of respondent (n=68)

Likely timing	18-44 years %	45-64 years %	65+ years %	Total %
Within 10 years	46.2	77.3	72.7	69.1
Within 30 years	30.8	4.5	15.2	14.7
Never	23.1	18.2	12.1	16.2
Total	100.0	100.0	100.0	100.0

Over two-thirds of respondents (68.1%) indicated they had no knowledge of firefighting infrastructure to combat wildfires in their municipality. Only 8 percent of respondents indicated that they had contacted authorities over the hazard rating for their residence. Perhaps low levels of personal knowledge of wildfires could be expected in a context where the management of the environmental hazard is placed in the hands of highly specialised mountain fire fighters. That argument is supported by findings that 98 percent of respondents approve of the cantonal forest fire fighting efforts, with 96 percent indicating support for those activities near their home.

6.3 Perceptions of management and planning for environmental risk

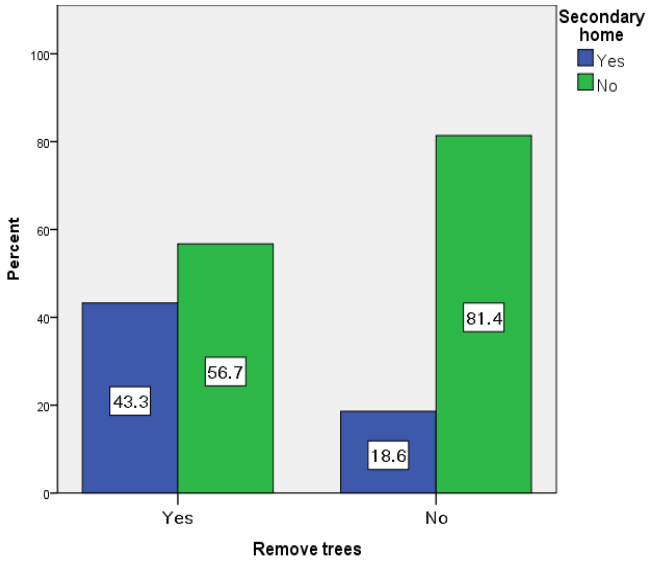
Respondents recognised the crucial function of the forest in stabilising steep slopes to protect settlements from gravitational hazards. Some 89 percent of respondents saw the forest as an intergenerational asset, recognising its bequest values, and over 72 percent of respondents were satisfied with the current extent of the forest. Concern was expressed about the lack of clearance of unsafe trees and dead wood, and many believed that the upkeep of forest areas should be made compulsory for landholders. Older respondents appeared more likely to perceive that tree-cover was too extensive. Respondents were asked the question: "Should clearing of trees be allowed close to settlements in some high risk wildfire areas to create buffer zones without forest?" A total of 61 percent of all respondents indicated that tree removal should be allowed in such areas (Table 5), with males significantly more likely than females to approve of the clearing of trees under such circumstances (χ^2 (n=109) = 5.353, p = .021).

Table 5: Respondents' support for tree clearing to create fire-safe buffer zones by gender (n=109)

Perception of tree clearance	Male %	Female %	Total %
Approve	67.5	43.8	60.6
Disapprove	32.5	56.3	39.4
Total	100.0	100.0	100.0

Further, 60 percent of respondents agreed that maintenance work on their property (vineyards, forested spaces and gardens) was important to reduce the risk of forest fires. A sense of shared responsibility is evident in the group of respondents who believed that the work they had carried out had a considerable impact on the wider landscape. Those same individuals who agreed that maintenance work was important to reduce risk were also more likely to believe in the effectiveness of the current fire policy that bans outdoor fires in periods of prolonged drought (χ^2 (n=111) = 19.273, p = .004). Respondents owning a *Monte*, a traditional transhumance residence, or another secondary home, were more likely to support clearance for buffer zones than those who did not (χ^2 (n=110) = 7.146, p = .008) (Figure 4). Most local wildfires start in the zone between the valley bottom and the *Monte*, halfway up the mountain slopes, and therefore disproportionately affect the former transhumance dwellings (Conedera and Pezzatti 2005; Conedera *et al.* 2015).

Figure 4: Preference for removing trees close to buildings as a function of owning a secondary home (n=110)



Over three-quarters of respondents considered current levels of rezoning for new housing development as excessive (Table 6). Those findings were supported by data showing that 85 percent of respondents held 'negative' or 'rather negative' feelings about new residential developments. At the same time, almost half of respondents saw a need for the establishment of stricter conservation areas, while one-third thought current levels were 'about right'.

Table 6: Percentage opinions of spatial planning priorities for the study area

	Percent			
Spatial planning priorities	**Too much already**	**About right**	**More needed**	**Total**
Set aside/ strict conservation areas (n=128)	18.8	34.4	46.8	100
Rezoning for new housing development (n=136)	78.7	9.6	11.8	100

Respondents were asked if there were any steps the federal, cantonal or municipal administration could take to enhance their feelings of personal safety by reducing the risk posed by natural hazards. Some 40 percent stated they did not think more work was needed to make them feel safer from environmental risk. A further 38 percent indicated that they were 'unsure' and were reluctant to speculate whether there was anything that could be done to enhance their safety. That said, respondents who did think that something could be done to enhance their protection (22% overall), provided 36 suggestions for enhanced protection from environmental hazards. Interestingly, some

42 percent of these suggestions involved thoughts on improving the management of forested landscapes, such as:

- *'Force landowners to keep the forest close to houses clean'*
- *'Make sure there is sufficient space between the forest and houses'*
- *'Eliminate unsafe trees'*
- *'Make the upkeep and cleaning of the forest compulsory, in such a way that the forest can breathe'*
- *'Fire-breaks'*

A substantial percentage of household participants indicated that they had some familiarity with the spatial planning system of Canton Ticino (Table 7), although only 3 percent believed that they held in-depth knowledge on the subject.

Table 7: Respondent familiarity with the planning system of Canton Ticino by age (n=135)

Familiarity	18-44 years %	45-64 years %	Over 65 years %	Total %
Very familiar	3.8	5.6	0.0	3.0
Some knowledge	84.6	79.6	90.9	85.2
Unsure	11.5	14.8	9.1	11.9
Total	100.0	100.0	100.0	100.0

On the other hand, most respondents felt that their views were not adequately reflected in current spatial planning outcomes (Table 8). Over three-quarters (79%) of respondents felt they had no influence over spatial planning issues concerning their local municipality or canton. Such results suggest that while traditional deliberative democratic channels are built into the planning system, the vast majority of respondents perceive they do not have a voice when it comes to local outcomes. There was a statistically significant difference based on the level of education of respondents, with tertiary educated respondents more likely to perceive that their opinions were influencing planning outcomes (57.7%) compared with respondents possessing only secondary education (55.1%) (FET (n= 124) = 7.294, p = .026).

Table 8: Perceptions of being heard on spatial planning issues concerning the local municipality or Canton by age (n=128)

Perception of being heard	18-44 years %	45-64 years %	Over 65 years %	Total %
Yes	17.4	12.0	30.9	21.1
No	82.6	88.0	69.1	78.9
Total	100.0	100.0	100.0	100.0

Although there are small numbers of respondents and the results are not significantly different (FTE (n=128) = 5.646, p = .055), 31 percent of respondents aged 65 years or more said that they do have a

voice, higher than the 12 percent of those aged 45-64 years or the 17 percent for those aged less than 45 years (Table 8). Perhaps elderly respondents are more supportive of current planning processes or, on the other hand, younger people's expectations of planning may be particularly high and are not being met. This result is interesting in the light of findings that show nearly three quarters (73.5%) of those aged 65 years or older gave a previous postcode from a non-Italian speaking region of Switzerland. Those respondents who gave their previous address as being from an non-Italian-speaking part of Switzerland were significantly more likely to feel they were being heard than those who gave a previous address from an Italian speaking part of Switzerland or abroad (χ^2 (n= 106) = 10.027, p = .002). In other words, the results suggest that respondents who were younger and from Ticino were less likely to support planning outcomes.

Respondents were asked to identify, from a list of approaches alternative to the official voting process, the ways in which they had been able to express their opinion on planning issues. Aside from voting in their official municipal ballots on particular issues, signing a petition and writing a letter to the municipality were the most popular alternative forms of direct engagement, while social media appeared to play only a small role (Table 9). The response rate to this question was low (n=70), perhaps because only those respondents who had participated in this manner chose to provide any details on how they expressed their opinion.

Table 9: Respondents' choice of alternative approaches to express opinions on planning issues in the study area (n=70)

	Number	Percent
Signing a petition	33	36.2
Letter to the Mayor/ municipality	25	27.5
Newspaper article	14	15.4
Social media	11	12.1
Rallies	8	8.8
	91*	100.0

*Multiple responses allowed

In summary, educational level and cultural/linguistic background are significant predictors of the sense of influence in relation to planning outcomes in Ticino. Once again, these results raise important issues for the effectiveness of the deliberative planning process, with influence apparently being reflective of certain groups within Canton Ticino, rather than a simple association with broad public participation in democratic processes.

7. Discussion

A high percentage (79%) of respondents felt they have 'no voice' in environmental planning outcomes concerning their local municipality/canton. This result was surprising, as the resident population of the study area have had numerous opportunities to participate in decision-making processes, and they are constantly being supported to learn about contemporary issues (Stanga 2017). Such a strong negative outcome could indicate a degree of despondency by respondents, who are expressing attitudes that might be expected in a population that views the democratic process as a mere 'window-dressing' exercise rather than a truly deliberative and representative process. Questions arise over the effectiveness of a political system in which people get to express their opinions and participate to an unparalleled degree, but feel that their participation does not influence outcomes (De'Arman 2020).

Survey results reveal a mismatch between respondents' perceptions of landscape amenity and ecological values, and their attitudes towards local risk mitigation approaches. On one hand there is a very high value attributed to the forested landscape and a wish for more trees and conservation (see also Conedera *et al.* 2015), but on the other there appears to be considerable support for clearing trees to immediately reduce risk. That situation, which is not unique to Ticino (Kalhor et al. 2018; Weber et al. 2019), suggests diverging priorities that may be impossible to reconcile fully within the one space. At the same time, current environmental management approaches could be creating a false sense of security in the population even as risks increase, while having the effect of 'losing the public' who clearly desire stricter conservation measures (Buchecker et al. 2003). Such an erosion of institutional trust (see Parkins and Mitchell 2005), seems to relate especially to immigrants from other Swiss cantons who often do not speak Italian and are more likely to misunderstand local risks, and as a result, may be particularly ignorant of the need for targeted interventions.

Even though we have analysed only a small sample, the results suggest that many decisions relating to planning for residents on the wildland-urban interface around Locarno do not fully incorporate local concerns – a result which appears to contrast with outcomes at the national level. Swiss planning generally enjoys great respect and ownership amongst the country's citizens, and partly as a result, democratic processes are granted higher political standing than in other countries (Kriesi and Trechsel 2008; Lendi 2012). The real strength of Swiss planning derives from its willingness to engage, anticipate potential problems and develop appropriate technical solutions in a process supported through the Swiss federal governance system (Frey 1994; Muggli 2013; Auer *et al.* 2014).

However, the results here would suggest that despite its clear merits, the system also generates potential procedural pitfalls for managing increasing natural hazards and urban expansion (Reinhard *et al.* 2005; Price *et al.* 2015). In this case, respondents may just be using the survey to voice their local grievances, but there is also the possibility that a perceived lack of citizen influence over environmental planning may originate from the fact that political, economic and technical experts increasingly dominate the Swiss policy agenda (Hersperger *et al.* 2014; Kruse and Pütz 2014). That is an issue affecting many democracies (Turner 2001), but in the Swiss case, the relative ease offered by popular referenda may be creating a disproportionate sense of citizen entitlement to be heard on local planning issues. Perhaps citizens' expectations are raised too high by the awareness that in such a robust democracy they should have a voice in all major decisions that affect them personally and collectively.

The highly deliberative Swiss system generates unique opportunities for citizen engagement, but simultaneously, institutional risks are created through an emphasis on the importance of participation (Stern and Baird 2015; Legacy 2017). The legitimacy of political engagement could erode if particular groups, such as many of the younger respondents from the survey in Ticino, do not see value in their involvement (see also Forester 1999). Although universal voting is not the aim of the Swiss deliberative democratic system, average participation rates in votes vary considerably. In two recent nation-wide examples, a referendum in 2018 saw participation reach 54 percent, while another in June 2018 drew only 25 percent of voters (FSO 2017; 2018). Participation for cantonal voting rarely exceeds 45 percent, with a 2017 vote on changes to the environmental protection law achieving 42 percent participation, while an earlier initiative to build a tunnel for a ring road attracting 36 percent of eligible voters (Repubblica e Cantone Ticino 2021). Dryzek (2009) argues that societies will need to avoid 'missing' or 'overhearing' the opinions of more vulnerable groups or, for that matter 'losing' groups that perceive that their involvement is meaningless if they wish to maintain effective democratic governance systems. In a society increasingly defined by risk (Beck 1986), that raises important questions about whether the deliberative planning processes will need to evolve to maintain its social licence.

Local planning in peri-urban spaces, such as the sample area, will need to increasingly incorporate the complex implications of environmental risk (Gallent and Shaw 2007; Von der Dunk *et al.* 2011; Nilsson *et al.* 2013). Within a deliberative democracy, the evolving situation must be understood by residents so they make an informed vote for their preferred outcomes. However, our results support a wider concern within the Swiss government (FOEN 2016) that residents are not fully

acknowledging that new risks are becoming more important with environmental change and so are not demanding changes in planning outcomes. That challenge is compounded by new levels of societal complexity. Particular groups within society, and especially those more vulnerable due to socio-economic or demographic circumstances, can miss out on opportunities to learn about risk (Fung and Wright 2003; Eriksen and Simon 2017). A specific planning focus on the wildland-urban interface, areas where socio-ecological risk will concentrate, could be one way to manage the unique, complex entanglement of urban growth, natural hazards and environmental values (Paveglio *et al.* 2009; Bardsley *et al.* 2015; Llausàs *et al.* 2016; Keller *et al.* 2020). Developing knowledge of residents' values and priorities for the management of such areas, while also educating about the new risk conditions, could be a vital step to informing such a planning approach.

Consultation does not necessarily imply deliberation (Dryzek 2000; Johnston 2014). Beck (2016) warns of the need for an evolution in democracy, as he asks how much environmental risk our democratic institutions can withstand before their basic tenets start to be undermined. As the public is not always keen to participate in democratic processes (Brady and Webb 2013), deliberative frameworks must be applied appropriately and be made attractive enough to encourage residents to engage by firstly ensuring that their actions will be meaningful, and secondly creating deliberative settings that are comfortable and facilitate exchange, communication and learning (Feng *et al.* 2018). It is vital that the challenge of emerging risks can be explained to society, so that people realise that responses to risk are difficult and expensive, and moreover, that there will always be residual risks that cannot be eliminated. The Swiss have the advantage of a traditional cultural heritage of risk management (Pfister 2009), where there is already trust in expert opinion, processes in collaborative learning, and an emphasis on place-based planning, seen as essential components of an effective approach (Champ *et al.* 2012; Wachinger *et al.* 2013). A middle path, where planning is open to diverse viewpoints and perspectives, but also capable of efficient and decisive action linked to scientific knowledge and technical expertise, has the potential to continue to generate a unique blend of both the instrumental and participatory planning approaches (Dryzek 1990; Allmendinger 2009; Inch 2015; Innes and Booher 2016). In doing so, a dialogue of risk reduction amongst the population, technical authorities and other government actors could be normalised, and a risk culture created and/or strengthened to enable sophisticated planning decisions (Hulme 2008).

8. Conclusion

Respondents from the upper slopes of the peri-urban area around Locarno highlighted a very strong attachment to their local forests. Yet, many elements are changing rapidly, including demographic and settlement patterns, forests and climate, which all contribute to changing levels of risk. Many respondents, and especially those originating from outside of Ticino, are not recognising the new levels of environmental risk, and seem to be more satisfied with current planning outcomes as a result. New forms of planning for risk would recognise that some decisions will need to mirror the perceptions of local residents, while others would continue to rely largely upon the technical understandings of risk. Such a comprehensive deliberative process would encourage residents of areas with high levels of environmental risk to engage more fully with relevant topics for their region. In turn, this has the potential to strengthen community ties. This is what Habermas (1984) refers to as the social room for public interaction, which is a crucial aspect in fostering positive risk mitigation behaviours across communities. However, local planning processes do not stand in a social and institutional vacuum; they must be considered within existing forms of democracy and norms of behaviour. One of the big unresolved questions in this field lies in the identification of mechanisms capable of translating the theory of deliberative planning into successful public participation. It may be that best practice deliberative planning will involve unique local combinations of participatory and technical inputs in decision-making at different levels of governance.

The Swiss democratic process creates knowledge and ownership of the planning system, thus resulting in degrees of approval in government rarely seen elsewhere. The success of that normalised process is reflected in the survey results, where 88 percent claim to have at least some knowledge of the cantonal planning system. The overwhelming satisfaction with fire-fighting authorities articulated in the survey also implies trust in the service that the government is providing. Yet, people have very high expectations of the planning system and most do not believe that their voices are being heard. As environmental risks become personal through impacts at local levels, the societal inability to enable democratic influences over planning outcomes appears to be generating challenges for governance. This article highlights the need for experimentation in this space to learn how to achieve better, real outcomes for society. In this case, even best-practice Swiss democratic processes struggle to guide knowledge and generate decisions that meet complex local needs and aspirations. As any reforms are initiated to respond to new levels of environmental risk, there will be limitations and failures and no-one has all the answers, so successful engagement

must be tied to learning processes. Perhaps the real finding is that a new level of uncertainty in relation to environmental risk must be made clear to all.

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